

- 4 (a) Fig. 4.1 shows two coherent sources P and Q producing waves. A wave detector is moved along line AB.

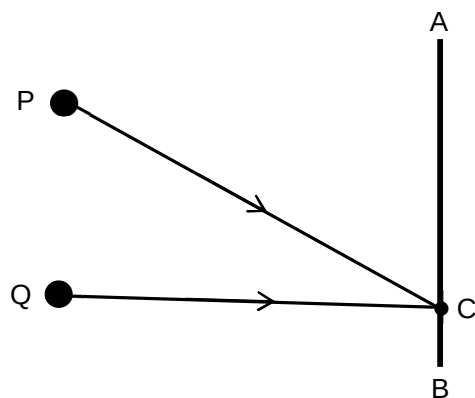


Fig. 4.1

- (i) Explain how interference fringes are formed along the line AB.

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 [2]

- (ii) The waves from P and Q meet at a point C in phase. Waves from P has an amplitude of 1.5 cm and intensity I . The resultant intensity at C is $3I$.

Calculate the amplitude of wave from Q.

amplitude = cm [2]

(b) The apparatus shown in Fig. 4.2 is used to produce an interference pattern on a screen.

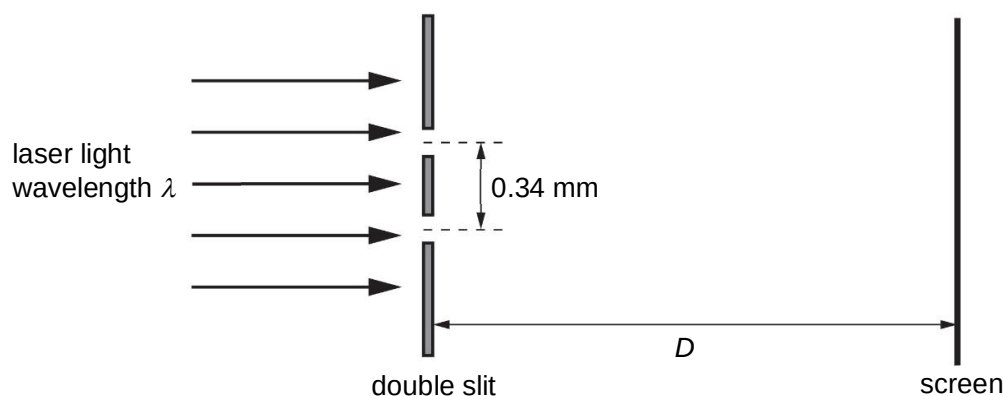


Fig. 4.2

Light of wavelength λ is incident on a double-slit. The slit separation is 0.34 mm. The separation between adjacent fringes is x . Fringes are viewed on a screen at distance D from the double-slit.

Distance D is varied from 2.0 m to 3.5 m. The variation with D of x is shown in Fig. 4.3.

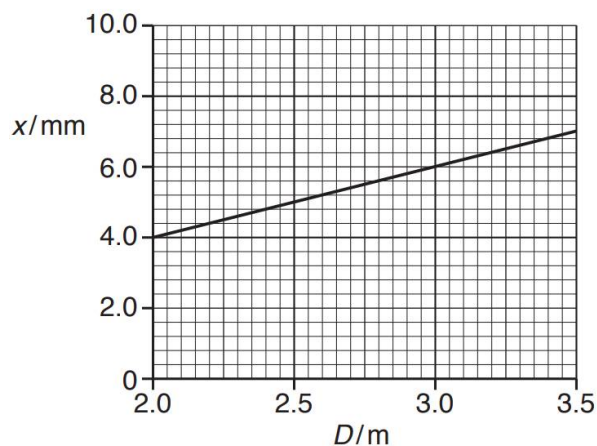


Fig. 4.3

(i) Use Fig. 4.3 to determine the wavelength λ .

$\lambda = \dots\dots\dots \text{ nm} \quad [3]$

(ii) The double slit is now replaced by another double slit with a larger slit separation.

On Fig. 4.3, sketch a possible line to show the variation with D of x for the fringes that are now produced. [1]

5 (a) A student decides to build a temperature probe and sets up the circuit using a